

CS22311-DATABASE MANAGEMENT SYSTEM LABORATORY

EX NO: 1.2

DATE:

DATA MANIPULATION LANGUAGE (DML) COMMANDS IN RDBMS

AIM:

To execute and verify the DML commands are the most frequently used SQL commands and is used to query and manipulate the existing database objects.

1. Insert values in a relation

Syntax: insert into tbname values(a,b,c),(d,e,f);

Query: insert into student

values(001,"abi",8.5,"cse"),(002,"john",8.6,"cse"),(003,"ram",8.2,"cse"),(004,"sam",9.0,"cse");

Output:

```
ad25104@adminincse05: ~
AD104>insert into student values(001,"abi",17,"cse"),(002,
"john",18,"cse"),(003,"ram",18,"cse"),(004,"sam",17,"cse")
;
Query OK, 4 rows affected (0.10 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

2. To display all the values

Syntax: select * from tbname;

Query: select * from student;

Output:

```
ad25104@adminincse0...
AD104>select * from student;
+-----+-----+-----+-----+
| rno | name | age | dept |
+-----+-----+-----+-----+
| 1 | abi | 17 | cse |
| 2 | john | 18 | cse |
| 3 | ram | 18 | cse |
| 4 | sam | 17 | cse |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

3. To update the values based on condition

Syntax: update tbname set attribute=value where condition ;

Query: update student set name="anu" where rno=1;

Output:

```
ad25104@admindcse05: ~  
AD104>update student set name="anu" where rno=1;  
Query OK, 1 row affected (0.03 sec)  
Rows matched: 1  Changed: 1  Warnings: 0  
  
AD104>select * from student;  
+-----+-----+-----+-----+  
| rno | name | age | dept |  
+-----+-----+-----+-----+  
| 1 | anu | 17 | cse |  
| 2 | john | 18 | cse |  
| 3 | ram | 18 | cse |  
| 4 | sam | 17 | cse |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)  
  
AD104>
```

4. To select a particular attribute from the table and display

Syntax: select attribname from tname;

Query: select name from student;

Output:

```
ad25104@admindcse05: ~  
AD104>select name from student;  
+-----+  
| name |  
+-----+  
| anu |  
| john |  
| ram |  
| sam |  
+-----+  
4 rows in set (0.00 sec)
```

5. To display more attributes from table

Syntax: select attrib1,attrib2,... from tname;

Query: select name,gpa from student;

Output:

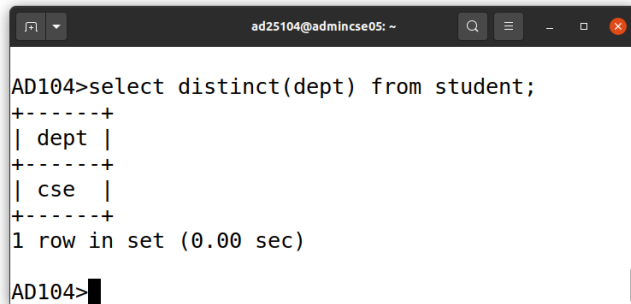
```
ad25104@admindcse05: ~  
AD104>select name,age from student;  
+-----+-----+  
| name | age |  
+-----+-----+  
| anu | 17 |  
| john | 18 |  
| ram | 18 |  
| sam | 17 |  
+-----+-----+  
4 rows in set (0.00 sec)
```

6. To select distinct values from a table

Syntax: select distinct (attribname) from tname;

Query: select distinct(department) from student;

Output:



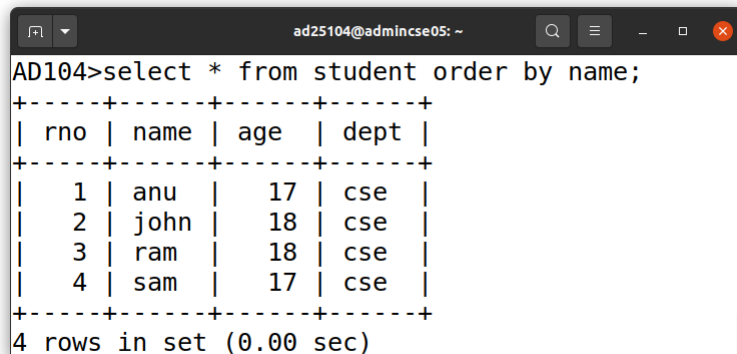
```
ad25104@adminincse05: ~  
AD104>select distinct(dept) from student;  
+-----+  
| dept |  
+-----+  
| cse  |  
+-----+  
1 row in set (0.00 sec)  
AD104>
```

7. To display records in ascending order

Syntax: select att1 from tname order by att2;

Query: select * from student order by name;

Output:



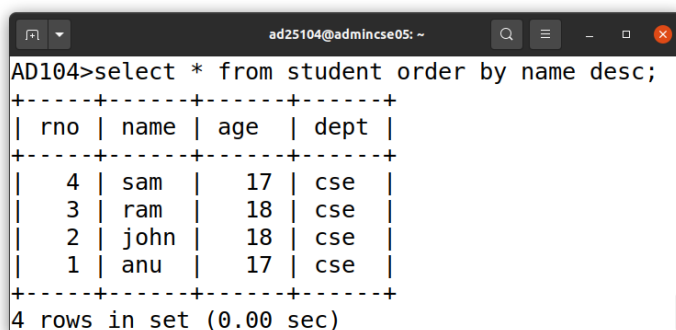
```
ad25104@adminincse05: ~  
AD104>select * from student order by name;  
+-----+-----+-----+-----+  
| rno | name | age | dept |  
+-----+-----+-----+-----+  
| 1 | anu | 17 | cse |  
| 2 | john | 18 | cse |  
| 3 | ram | 18 | cse |  
| 4 | sam | 17 | cse |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

8. To display records in descending order

Syntax: select att1 from tname order by att2 desc;

Query: select * from student order by name desc;

Output:



```
ad25104@adminincse05: ~  
AD104>select * from student order by name desc;  
+-----+-----+-----+-----+  
| rno | name | age | dept |  
+-----+-----+-----+-----+  
| 4 | sam | 17 | cse |  
| 3 | ram | 18 | cse |  
| 2 | john | 18 | cse |  
| 1 | anu | 17 | cse |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

9. To select attributes with and/or condition

Syntax: select attribname from tname where condition1 and/or condition2;

Query: select name from student where rno=3 and department="cse";

Output:

```
ad25104@admincse05: ~  
AD104>select name from student where rno=3  
+-----+="cse";  
| name |  
+-----+  
| ram  |  
+-----+  
1 row in set (0.00 sec)  
AD104>
```

10. Insert particular value

Syntax: insert into tname (attribname) values ("xxx");

Query: insert into student(rno) values("5");

Output:

```
ad25104@admincse05: ~  
AD104>insert into student(rno) values("5");  
Query OK, 1 row affected (0.06 sec)  
AD104>select * from student;  
+-----+-----+-----+-----+  
| rno | name | age | dept |  
+-----+-----+-----+-----+  
| 1 | anu | 17 | cse |  
| 2 | john | 18 | cse |  
| 3 | ram | 18 | cse |  
| 4 | sam | 17 | cse |  
| 5 | NULL | NULL | NULL |  
+-----+-----+-----+-----+  
5 rows in set (0.00 sec)  
AD104>
```

11. Delete a record

Syntax: delete from tname where condition;

Query: delete from student where rno="5";

Output:

```
ad25104@admincse05: ~  
AD104>delete from student where rno="5";  
Query OK, 1 row affected (0.05 sec)  
AD104>select * from student;  
+-----+-----+-----+-----+  
| rno | name | age | dept |  
+-----+-----+-----+-----+  
| 1 | anu | 17 | cse |  
| 2 | john | 18 | cse |  
| 3 | ram | 18 | cse |  
| 4 | sam | 17 | cse |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)  
AD104>
```

12. Delete a constraint

Syntax: alter table tname drop primary key;

Query: alter table student drop primary key;

Output:

```
ad25104@admincse05: ~  
AD104>ALTER TABLE student  
-> DROP PRIMARY KEY;  
Query OK, 4 rows affected (0.83 sec)  
Records: 4 Duplicates: 0 Warnings: 0  
  
AD104>desc student;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| rno   | int           | NO   |     | NULL    |       |  
| name  | varchar(30)   | YES  | UNI | NULL    |       |  
| age   | int           | YES  |     | NULL    |       |  
| dept  | varchar(10)   | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.00 sec)  
  
AD104>█
```

RESULT:

The required data manipulation language commands were successfully executed .